# Relations and functions

**1**

| $x$ | $y$ |
|---|---|
| 2 | 2 |
| 4 | 4 |
| 6 | 3 |
| 7 | 5 |

**2**

**a** True **b** False **c** True **d** True

**e** False **f** False **g** False **h** False

**i** True **j** False **k** False **l** False

**3**

**a** Both a relation and a function **b** Relation only

**c** Relation only **d** Relation only

**e** Both a relation and a function **f** Both a relation and a function

**g** Relation only **h** Both a relation and a function

**i** Both a relation and a function **j** Relation only

**4**

**a** $16, 8, -15, -20$ **b** $14, 28, -28$ **c** Yes

**5**

**a** Both a function and a relation **b** Relation

**c** Relation **d** Both a function and a relation

**e** Relation

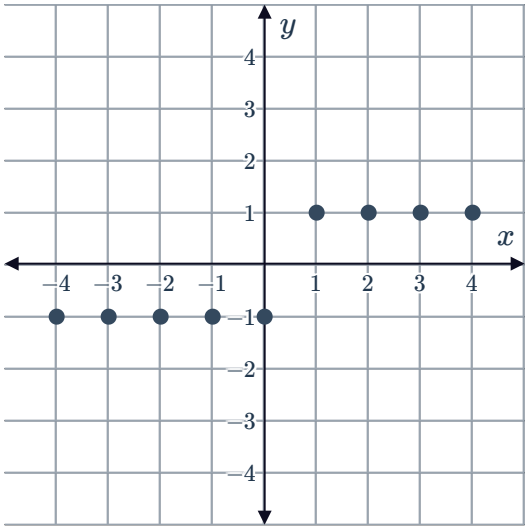
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**6**

**a**

| $x$ | $-4$ | $-3$ | $-2$ | $-1$ | $0$ | $1$ | $2$ | $3$ | $4$ |
|---|---|---|---|---|---|---|---|---|---|
| $y$ | $-1$ | $-1$ | $-1$ | $-1$ | $-1$ | $1$ | $1$ | $1$ | $1$ |

**b**



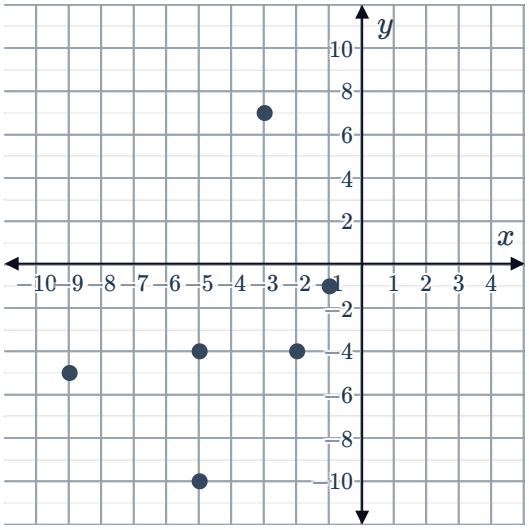

**c** Yes

**7**

**a** Yes **b** Yes **c** Yes **d** No

**8**

**a**




**b** No

**c** Either $(-5, -10)$ or $(-5, -4)$

**9**

**a** No such value of $k$ exists **b** $8, 6, 1$

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## Function notation

**10**
**a** $46$ **b** $-189$ **c** $\frac{33}{4}$ **d** $87$

**11**
**a** $g\left(7\right)=-6$ **b** $f\left(x\right)=\frac{1}{4}x-2$

**12**
**a** $2$ **b** $0$ **c** $1$

**13**
**a** $1$ **b** Yes

**14**
**a** $12$ **b** $-16$

**15**
**a** $8$ **b** $-13$

**16**
**a** $\frac{35}{4}$ **b** $-7$

**17**
**a** $1$ **b** $-\frac{13}{5}$

**18**
**a** $68$ **b** $-4$

**19** $\frac{9}{2}$

**20**
**a** $0$ **b** $3$ **c** $\frac{91}{25}$

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**21**

**a**

**i** $-48$ **ii** $15$ **iii** $-49$

**b** $x = \pm 7.81$

**22**

**a** $a^2 + 8a$

**b** $b^2 + 8b$

**23**

**a** $-4$

**b** $-\dfrac{121}{32}$

**24**

**a** $0$ **b** $2.67$ **c** $80.99$

**25** $-46$

**26**

**a** $26$ **b** $52$ **c** $32$

**27**

**a** $\sqrt{128}$ **b** $12$ **c** $\sqrt{63}$ **d** $\sqrt{142}$

**28**

**a** $12$

**b** $m^2 + 8$

**29**

**a** $39$

**b** $-25$

**c** $a^3 - 8a + 7$

**d** $b^3 - 8b + 7$

**e** $(a+b)^3 - 8(a+b) + 7$

**f** No

**30**

**a**

**i** $f(x) = 2 - \dfrac{1}{3}x$

**ii** $1$

**b**

**i** $f(x) = \dfrac{6}{5}x + \dfrac{7}{5}$

**ii** $5$

**c**

**i** $f(x) = 3 - x - 6x^2$

**ii** $-54$

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**31**

**a**

**i** $f(x) = \dfrac{7 - 5x}{-3}$

**ii** $1$

**b**

**i** $f(x) = 4x^2 + x + 5$

**ii** $23$

**32**

**a** $f(x) = \dfrac{1}{4}x - 2$

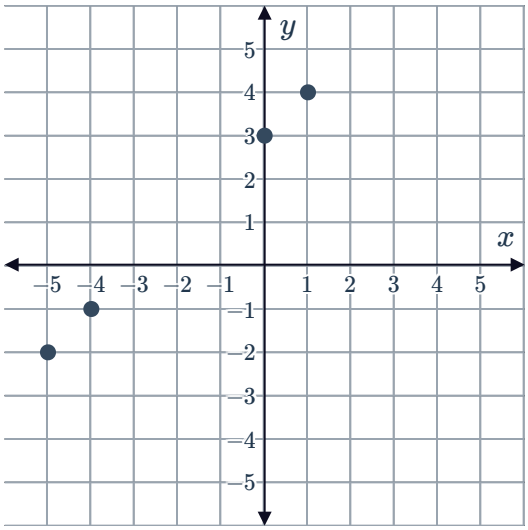
**b** $1$

**33**

**a**

| $x$ | $-5$ | $-4$ | $0$ | $1$ |
|---|---|---|---|---|
| $f(x)$ | $-2$ | $-1$ | $3$ | $4$ |

**b**

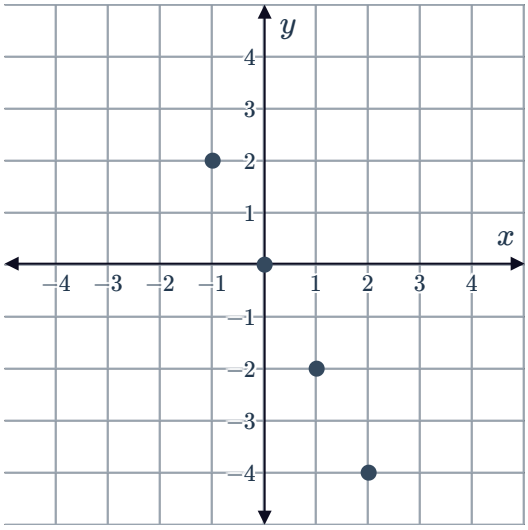



**34**

**a**

| $x$ | $-1$ | $0$ | $1$ | $2$ |
|---|---|---|---|---|
| $f(x)$ | $2$ | $0$ | $-2$ | $-4$ |

**b**



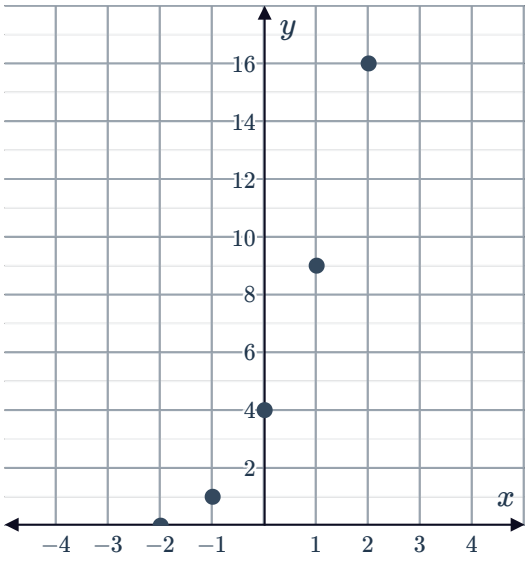

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**35**

**a**

| $x$ | $-2$ | $-1$ | $0$ | $1$ | $2$ |
|---|---|---|---|---|---|
| $f(x)$ | $0$ | $1$ | $4$ | $9$ | $16$ |

**b**

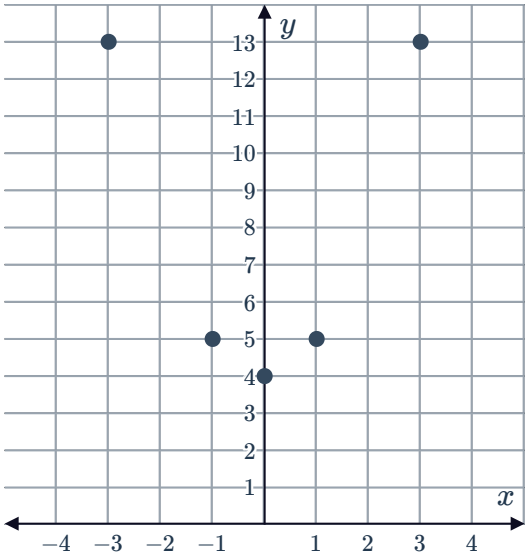



**36**

**a**

| $x$ | $-3$ | $-1$ | $0$ | $1$ | $3$ |
|---|---|---|---|---|---|
| $f(x)$ | $13$ | $5$ | $4$ | $5$ | $13$ |

**b**



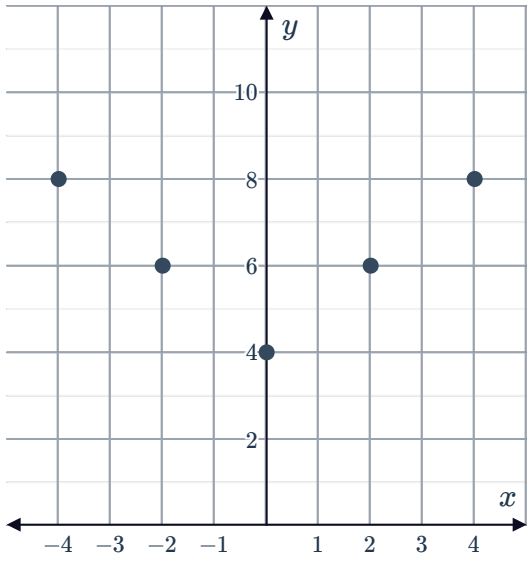

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**37**

**a**

| $x$ | $-4$ | $-2$ | $0$ | $2$ | $4$ |
|---|---|---|---|---|---|
| $f(x)$ | $8$ | $6$ | $4$ | $6$ | $8$ |

**b**




## Applications

**38**

**a**

| Call length (minutes) | International Call cost |
|---|---|
| 1 | $3.50 |
| 2 | $6.40 |
| 3 | $9.30 |
| 4 | $12.20 |
| 5 | $15.10 |

**b** Yes

**39**

**a**

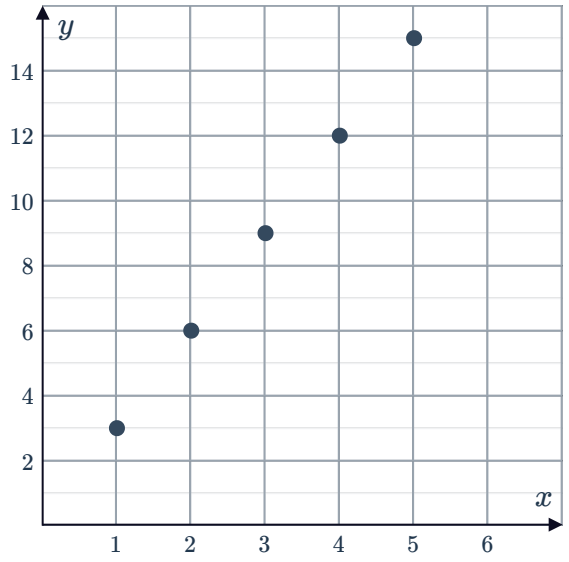
| Number of T-shirts | Total Cost (dollars) |
|---|---|
| 1 | $19 |
| 2 | $38 |
| 3 | $38 |
| 4 | $57 |
| 5 | $76 |

**b** Yes

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**40**

**a**




**b** Yes

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